

Section A

[25 marks]

This section consists of 3 questions. Answer *all* questions.

1. Evaluate each of the following limits, if exist.

(a) $\lim_{x \rightarrow 3} \frac{2x - 6}{3 + 2x - x^2}$

[3 marks]

(b) $\lim_{x \rightarrow 3^-} \frac{x^2 - 9}{|x - 3|}$

[3 marks]

(c) $\lim_{x \rightarrow -\infty} \frac{x + 2}{\sqrt{4x^2 + 1}}$

[3 marks]

2. (a) Given that $f(x) = \sqrt{2x + 3}$. By using the first principle, find $f'(x)$.

[5 marks]

(b) Given that $y = e^{3x} \ln(2x)$, find $\frac{dy}{dx}$.

[4 marks]

3. A curve has the equation $y = \frac{1}{3}x^3 - 2x^2 + 3x + 8$. Find the coordinates of the stationary point and determine whether this is a maximum or minimum.

[7 marks]

Section B

[75 marks]

This section consists of 7 questions. Answer *all* questions.

1. The complex number Z_1 and Z_2 are given by $Z_1 = 3 + 2i$ and $Z_2 = 1 - 3i$.

Find $Z_3 = \frac{Z_2}{Z_1}$ in the form of $a + bi$.

Hence, find $\text{Arg}(Z_3)$.

[5 marks]

2. (a) Solve $e^x - 12e^{-x} - 1 = 0$.

[5 marks]

(b) By writing $|x| \geq a$ in the equivalent for $x \geq a$ or $x \leq -a$, determine the solution set for

$$\left| \frac{x-3}{2x-1} \right| \geq \frac{1}{2}$$

[6 marks]

3. The function f and g are given by $f(x) = e^{2x} + 1$ and $g(x) = \frac{1}{2} \ln(x-1)$.

(a) Show that $f(x)$ is a one-to-one function.

[2 marks]

(b) Determine the inverse function of $g(x)$.

Hence, state the relationship between $f(x)$ and $g(x)$.

[4 marks]

(c) Find the function $h(x)$ if $(f \circ h)(x) = e^{4x^2} + 1$.

[3 marks]

4. Given the function $f(x) = \frac{1}{2x-3}$.

(a) Find the domain and range of f .

[2 marks]

(b) Show that $f(x)$ is a one-to-one function by using algebraic approach.

Hence, find $f^{-1}(x)$.

[5 marks]

(c) Hence, sketch the graph f and f^{-1} on the same axes. Label your graph completely.

[3 marks]

5. (a) Evaluate $\lim_{x \rightarrow 4} \frac{2 - \sqrt{x}}{4 - x}$.

[3 marks]

(b) Given $f(x) = \frac{2x}{(x+1)(x-2)}$.

By finding the limit at appropriate points, obtain the horizontal and vertical asymptotes.

[6 marks]

(c) Function f is as follows:

$$f(x) = \begin{cases} e^x, & x \geq 0 \\ 1 - 9x^2, & x < 0 \end{cases}$$

(i) Find $\lim_{x \rightarrow 0} f(x)$.

[3 marks]

(ii) Determine whether f is continuous at $x = 0$. Give your reason.

[3 marks]

6. (a) If $xy = 2(x - y)^2$, find the following values at $\left(1, \frac{1}{2}\right)$

(i) $\frac{dy}{dx}$

[5 marks]

(ii) $\frac{d^2y}{dx^2}$

[4 marks]

(b) Given $x = \cos 2t$ and $y = 2t + \sin^2 t$, show that $\frac{dy}{dx} = -\operatorname{cosec} 2t - \frac{1}{2}$ and

evaluate $\frac{d^2y}{dx^2}$ when $t = \frac{\pi}{8}$.

[7 marks]

7. A cylindrical container has one end open and the other end closed. It has a circular base of radius r cm. Given that the total surface area of container is 300π cm².

(a) Show that the volume of the container is $V = 150\pi r - \frac{\pi r^3}{2}$ cm³.

[4 marks]

(b) Find the maximum volume of container, V .

[5 marks]

END OF QUESTION PAPER